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[22AS52]

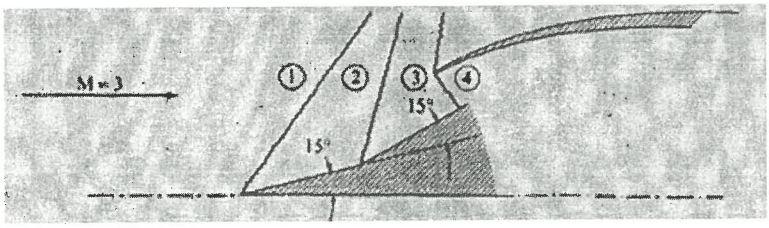
FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

AIRCRAFT PROPULSION

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Missing data may be suitably assumed.
ii. Use of gas table is permitted

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	What is by pass ratio of a turbofan engine? Give the typical bypass ratio range for the modern day engines.	L1	CO1	**02**
	b.	List the functions of an aircraft engine inlet.	L1	CO 2	**02**
	c.	Name the methods used for thrust reversing in an aircraft.	L1	CO 3	**02**
	d.	Define flame stabilization.	L1	CO 4	**02**
	e.	Differentiate between impulse and reaction turbine.	L2	CO 5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Derive an expression for thrust produced by a jet engine.	L3	CO 1	**05**
		OR			
	b.	With neat sketch, Explain the working of Turbofan engine with its advantages and disadvantages.	L2	CO 1	**05**
3.	**a.**	Determine the total pressure recovery for a supersonic inlet with operating Mach number of 3 at the sea level conditions. 	L3	CO 2	**05**
		OR			
	b.	Describe Shock swallowing technique by area variation in the supersonic inlet with neat sketch.	L2	CO 2	**05**
4.	**a.**	Describe the classification and function of an aircraft nozzle.	L2	CO 3	**05**
		OR			
	b.	Explain the working principle of a Chevron nozzle with a neat sketch.	L2	CO 3	**05**
5.	**a.**	Illustrate the construction and application of stability loop.	L2	CO 4	**05**
		OR			
	b.	Describe the pressure loss in the combustion chamber.	L2	CO 4	**05**

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6.	a.	Illustrate the working of centrifugal compressor.	L2	CO 5	**05**
		OR			
	b.	Explain any three performance coefficients of axial compressor.	L2	CO 5	**05**
7.	a.	Illustrate the liquid cooling methods of turbine.	L2	CO 6	**05**
		OR			
	b.	Mention the three dimensionless parameters used in aerodynamic design of a turbine stage and explain them.	L2	CO 6	**05**
		PART-C	**6 X 10=60**		
8.	a.	A McDonnell F4C airplane is powered by two General Electric J79-GE-15 turbojet engines. F4C has the following characteristics: wing area S=49 m^2, engine inlet area A_1=0.6 m^2, cruise Mach number M_0=0.82, and flight altitude of 9100 m. The drag coefficient of the aircraft under these conditions is 0.045. Determine: (a) The net thrust of the engine (b) The mass flow rate through the engine (c) The gross thrust of the engine Then, if the exhaust pressure $P_e = P_0$ (d) Estimate the exhaust velocity V_e assuming that the turbine exit stagnation temperature (which is the nozzle entry stagnation temperature) as T_{0e}=1005.4 K (e) Estimate the exit static temperature T_e (f) The exit Mach number M_e	L3	CO 1	**10**
		OR			
	b.	An ideal Otto cycle has a compression ratio of 8. At the beginning of the compression process, air is at 95 kPa and 27°C, and 750 kJ/kg of heat is transferred to air during the constant-volume heat-addition process. Taking constant specific heats at room temperature., determine (a) the pressure and temperature at the end of the heat- addition process, (b) the network output, (c) the thermal efficiency, and (d) the mean effective pressure for the cycle.	L3	CO 1	**10**
9.	a.	Illustrative the Successive steps in the acceleration and over speeding of a one-dimensional supersonic inlet.	L2	CO 2	**10**
		OR			
	b.	Explain the flow features in the subsonic inlet with neat sketch.	L2	CO 2	**10**

10.	**a.**	Describe the three modes of operation of a C-D nozzle with the help of pressure variation along the length of the nozzle.	L2	CO 3	**10**
		OR			
	b.	Illustrate the working of afterburner with neat sketch along with modified T-S diagram.	L2	CO 3	**10**
11.	**a.**	Illustrate the methods of flame stabilization with suitable sketch.	L2	CO 4	**10**
		OR			
	b.	Differentiate between the can type and annular type combustors with the help of schematic sketches.	L2	CO 4	**10**
12.	**a.**	Explain the relationship between the degree of reaction and the aerodynamic design of an axial compressor. How does it affect the pressure rise across the rotor and stator?	L2	CO 5	**10**
		OR			
	b.	Discuss the various types of compressors and their typical applications, focusing on the advantages and limitations of each type.	L2	CO 5	**10**
13.	**a.**	Describe the different spool configurations used in turbine systems. How do they affect the performance and operational flexibility of the engine?	L2	CO 6	**10**
		OR			
	b.	What is turbine-compressor matching, and why is it important in the design of gas turbine engines? Discuss the factors that need to be considered for effective matching.	L2	CO 6	**10**